SIMULATED DATA: illustrative example of the old measuring method, not measured values

Hand-measured vs model values for 200 synthetic sheets, per trait

Actual vs model

dashed line = perfect agreement

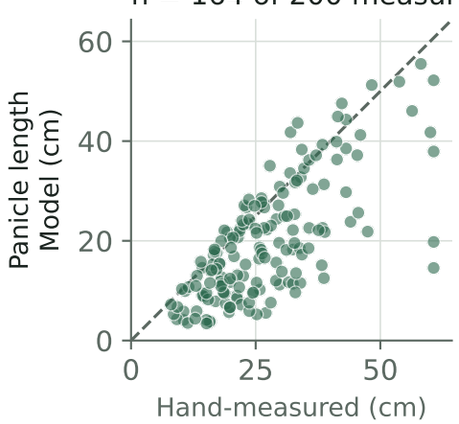
Error distribution

green band = within ±20%; orange = mean bias

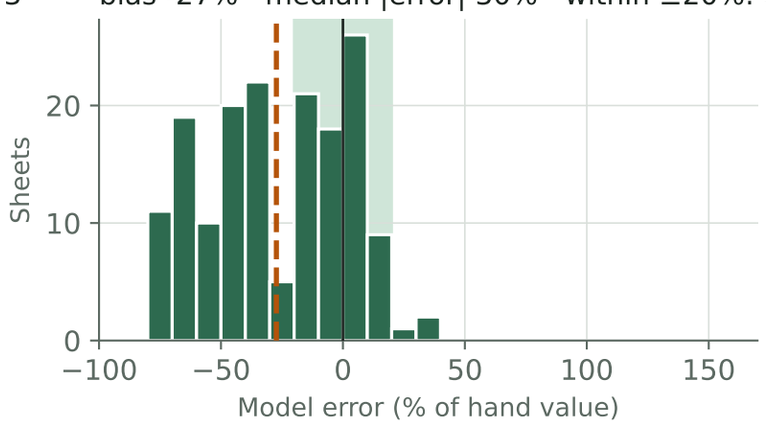
Bias and agreement

Bland–Altman: orange = mean and ±1.96 SD

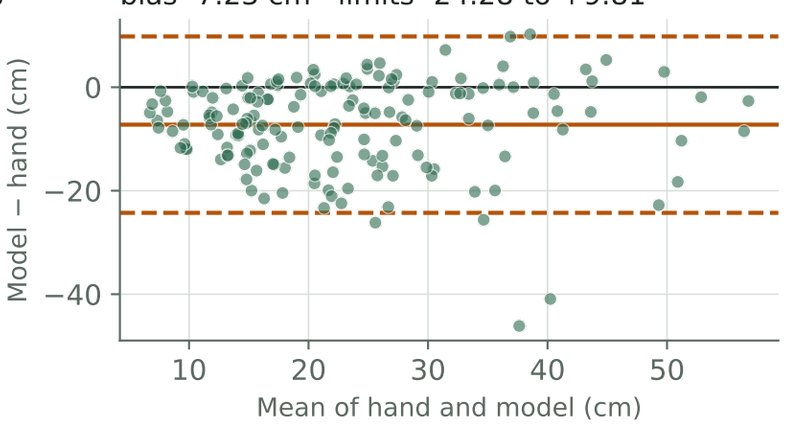
n = 164 of 200 measured (82%) r = 0.73



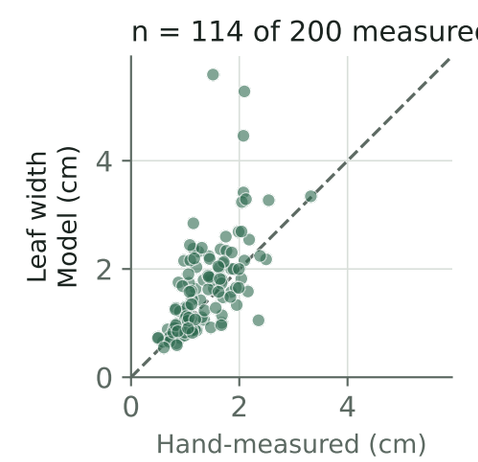
bias -27% median |error| 30% within ±20%: 45%



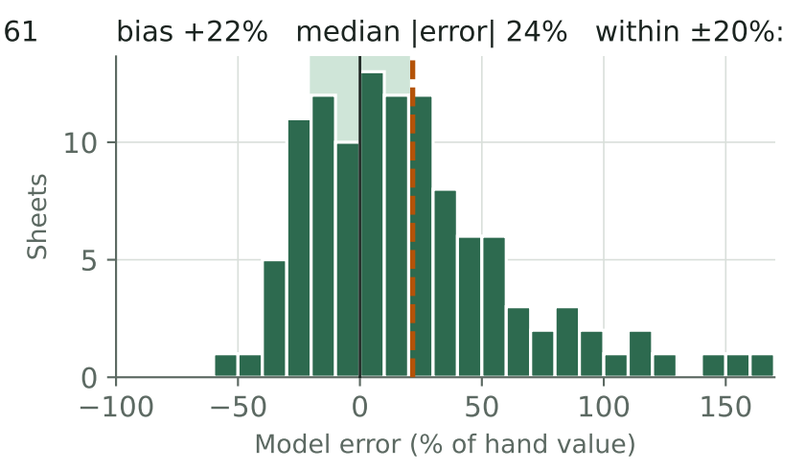
bias -7.23 cm limits -24.28 to +9.81



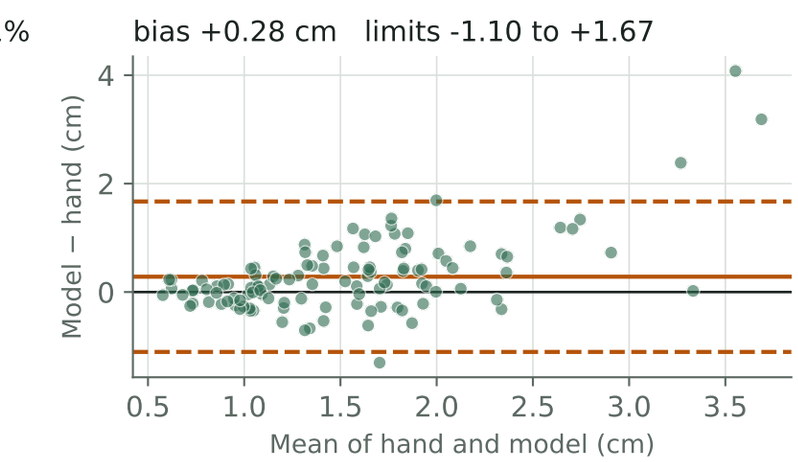
n = 114 of 200 measured (57%) r = 0.61



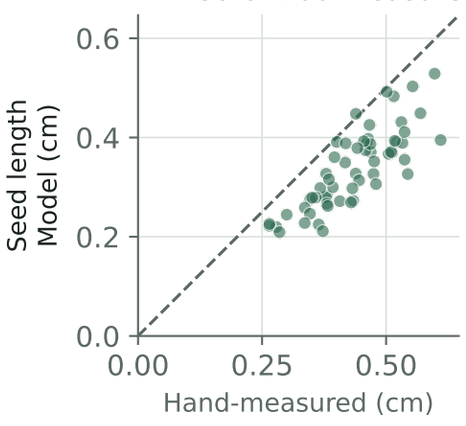
bias +22% median |error| 24% within ±20%: 41%



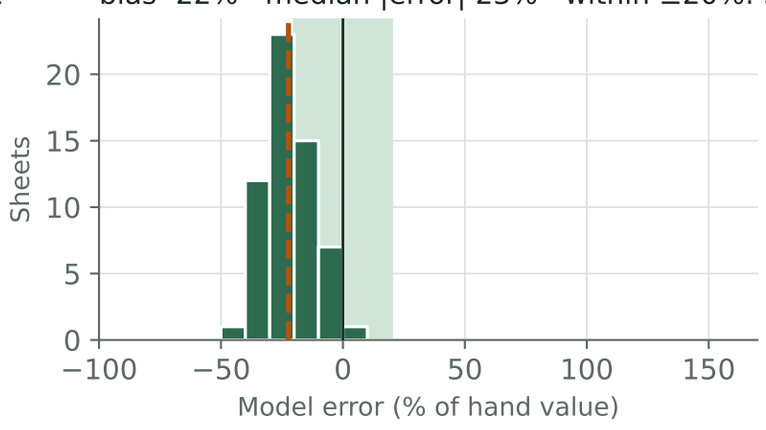
bias +0.28 cm limits -1.10 to +1.67



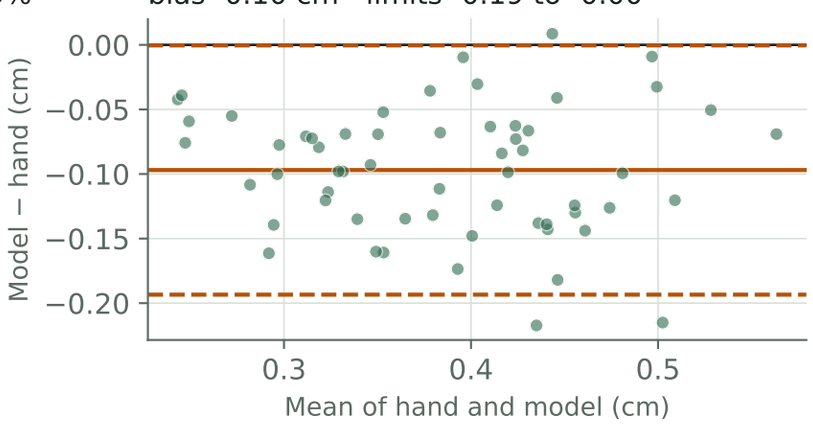
n = 59 of 200 measured (30%) r = 0.82



bias -22% median |error| 23% within ±20%: 39%



bias -0.10 cm limits -0.19 to -0.00



Simulated (n = 200 synthetic sheets per trait, random seed 2026); no real sheets were measured for this figure. Panicle and seed behaviour follow the failure pattern observed in a real 10-sheet benchmark of the old LLM-only method (panicle: bias −33%, median error 25%; seed: 30% measured, bias −22%). Leaf behaviour is assumed. Hand-measured values are drawn from the range of the lab's 163 real hand measurements of Megathyrsus maximus.